

1. BACKGROUND

The main thing aimed at here is to get a natural measure on V -variable fractals which is preserved under an appropriate transfer operator, and which is a Gibbs measure in an appropriate sense.

The following uses a little of the notation from the Advances paper and notation from your handwritten notes.

We are working with weighted IFSs, such as

$$F = (f_1^F, f_2^F, f_3^F; w_1^F, w_2^F, w_3^F), \quad G = (f_1^G, \dots, f_6^G; w_1^G, \dots, w_6^G),$$

where F and G correspond to $SG(2)$ and $SG(3)$, with weights which typically are $w_i^F = (r_i^F)^\alpha$ where $r_i^F = \text{Lip } f_i^F$, etc., and α is to be found from a Ruelle-Perron-Frobenius type theorem which we need to know/show works in our context.

It seems the appropriate space consists of pairs

$$\Omega_V^* \otimes \Sigma^V := ((\omega^{(1)}, \dots, \omega^{(V)}), (\sigma^{(1)}, \dots, \sigma^{(V)}))$$

where $(\omega^{(1)}, \dots, \omega^{(V)})$ is a V -variable V -tuple and $\sigma^{(i)} \in \Sigma(\omega^{(i)})$.

In order to define the transfer operator on functions it seems necessary to work with V -tuples. Note also that one needs a path (i.e. point) in each $\omega^{(i)}$. This also seems necessary — working with an address a as in your handwritten notes does not seem to contain enough information. Think about what happens in the trivial case $V = 1$ and just one IFS, which is the usual deterministic setting.

The transfer operator is then the natural one:

$$(1) \quad \begin{aligned} & T((\omega^{(1)}, \dots, \omega^{(V)}), (\sigma^{(1)}, \dots, \sigma^{(V)})) \\ &= \left((T_{\sigma_1^{(1)}} \omega^{(1)}, \dots, T_{\sigma_1^{(V)}} \omega^{(1)}), (T\sigma^{(1)}, \dots, T\sigma^{(V)}) \right), \end{aligned}$$

using your notation. T does preserve the class defined above.

The definition of the transfer operator $\mathcal{L}$ on functions needed a lot of thought for me. The approach used by Patschke does not work — the problem is that it does not lead to a natural transfer operator on functions with good properties in the V -variable case.

But what Patschke does is equivalent to the following idea in the random recursive case.

First note that in the deterministic case,

$$(\mathcal{L}h)(\sigma) = \sum_i w_i h(i\sigma) = \sum_{\{\tau: T\tau=\sigma\}} w_{\tau_1} h(\tau).$$

This can also be interpreted, at least in the case that the w_i are probabilities, as

$$(\mathcal{L}h)(\sigma) = \mathbb{E}^x h(X(\sigma)),$$

where X is a random variable starting at σ and proceeding to $i\sigma$ with probability w_i .

In the random recursive case, motivated by the above, it seems natural to define

$$(2) \quad (\mathcal{L}h)(\omega, \sigma) = \mathbb{E}_{\tilde{\omega}} \sum_{i=1}^{N(\tilde{\omega})} \left(w_i(\tilde{\omega}) h(\tilde{\omega}, i\sigma) \mid T_i \tilde{\omega} = \omega \right).$$

In other words, we are integrating (weighting) with respect to those $\tilde{\omega}$ subject to the conditional probability that at least one of the level one subtrees of $\tilde{\omega}$ is the same as ω , and then weighting again with respect to the relevant w_i .

By using independence, this is equivalent to the form Patschke uses, and he thus avoids using conditional probabilities. (I should check this carefully).

But in the V -variable case, (2) and the version used by Patschke are not equivalent — the $V = 1$ case is an example.

2. TRANSFER OPERATOR ON FUNCTIONS

From (1) and (2) one is led first to a natural definition for $\mathcal{L}_V$ in the $V = 1$ case and then for $\mathcal{L}_V$ in the case of arbitrary V .

To motivate this, consider Figure 1.

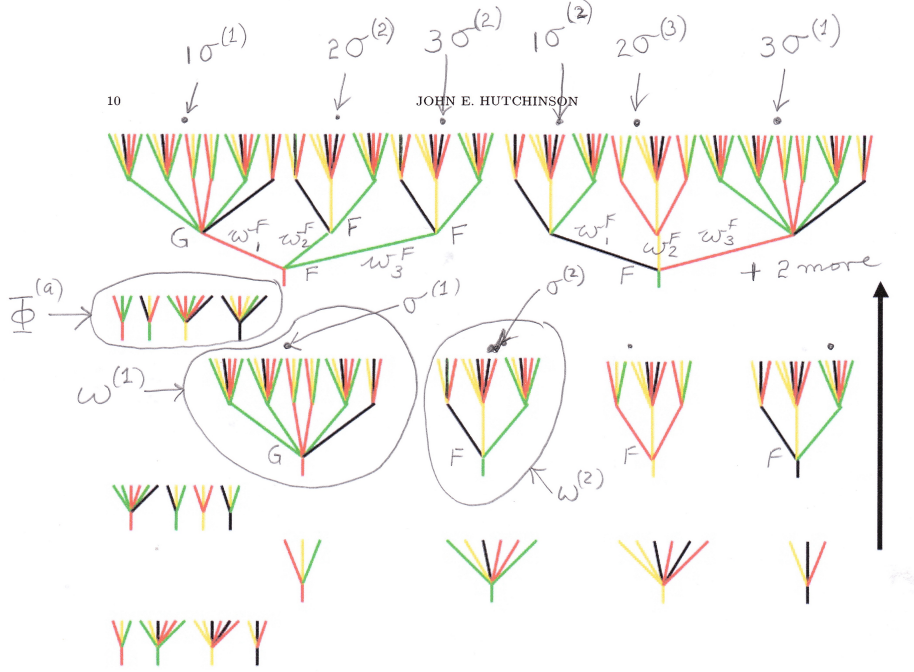


FIGURE 1. Transfer Operator

The third line from the top is $(\omega^{(1)}, \dots, \omega^{(4)})$ and the top line is

$$(\tilde{\omega}^{(1)}, \dots, \tilde{\omega}^{(4)}) = \Phi^a(\omega^{(1)}, \dots, \omega^{(4)}).$$

An example of paths $\sigma^{(i)}$ is shown in the third line from the top, and some of the paths $j\sigma^{(i)}$ are shown in the top line.

The operator Φ^a is given by the 4 small trees in the second line from the top. In more standard notation,

$$(3) \quad \Phi^a = \begin{bmatrix} F & 1 & 2 & 2 & * & * & * \\ F & 4 & 3 & 1 & * & * & * \\ G & 2 & 1 & 1 & 2 & 1 & 4 \\ G & 4 & 3 & 1 & 3 & 2 & 4 \end{bmatrix}$$

For the box notation, see Figure 2.

In order to motivate the general definition of the transfer operator on functions, first consider the case $V = 1$. Then $a \in \mathcal{A}_V$ is completely described by choice of an IFS, and so in our test case with just two IFSs, a corresponds to either F or G .

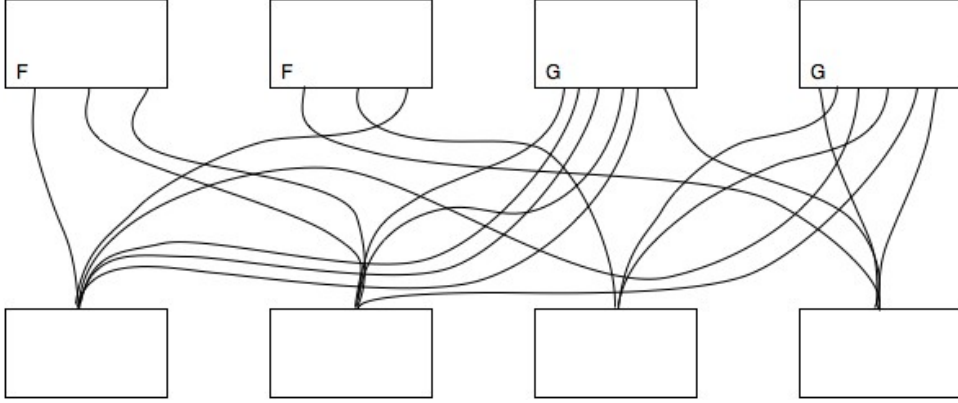


FIGURE 2. The bottom row of boxes contain $\omega^{(1)}, \dots, \omega^{(4)}$. The operator Φ^a is then applied to give $\tilde{\omega}^{(1)}, \dots, \tilde{\omega}^{(4)}$, which are placed in the top row of boxes. The operator Φ^a here is the same as in Figure 1.

In this test case suppose that the initial probability P on IFSs is given by $P = \{p_F, p_G\}$. Then (2) is equivalent to the definition (this needs explanation)

$$(\mathcal{L}_1 h)(\omega, \sigma) = p_F \sum_{i=1}^3 w_i^F h(F(\omega), i\sigma) + p_G \sum_{i=1}^6 w_i^G h(G(\omega), i\sigma).$$

In the general $V = 1$ case (needs similar explanation)

$$\begin{aligned} (\mathcal{L}_1 h)(\omega, \sigma) &= \int \left(\sum_{i \in N(F^\lambda)} w_i^\lambda h(F^\lambda(\omega), i\sigma) \right) dP(\lambda) \\ &= \int \left(\sum_{i \in N(I(a))} w_i^{I(a)} h(\Phi^a(\omega), i\sigma) \right) dP_1(a), \end{aligned}$$

where $N(F^\lambda)$ is the number of functions in the IFS F^λ and $N(I^a)$ is the number of functions in the IFS F^{I^a} corresponding to a . Note that Φ^a corresponds to a single IFS F^λ , and conversely. The initial probability P on Λ is the same as the probability P_1 on $\mathcal{A}_1$.

Now consider the case for general V . We define a transfer operator acting on functions $h : \Omega_V^* \circ \Sigma^* \rightarrow \mathbb{R}$.

For $a \in \mathcal{A}_V$ let

$$\Gamma^a = \{1, \dots, N(I^a(1))\} \times \dots \times \{1, \dots, N(I^a(V))\}$$

index the set of “allowable” V -tuples of level one branches. In Figure 2 this means

$$\Gamma^a = \{1, 2, 3\} \times \{1, 2, 3\} \times \{1, \dots, 6\} \times \{1, \dots, 6\}.$$

For $\gamma \in \Gamma^a$ let

$$w_\gamma^a = w_{\gamma(1)}^{I^a(1)} \times \dots \times w_{\gamma(V)}^{I^a(V)}$$

be the corresponding product of weights. Note that if the w_i^F are probabilities then $\sum_{\gamma \in \Gamma^a} w_\gamma^a = 1$.

Now once again, (2) is here equivalent (this needs explanation) to the following definition

$$(4) \quad (\mathcal{L}_V h) \left((\omega^{(1)}, \dots, \omega^{(V)}), (\sigma^{(1)}, \dots, \sigma^{(V)}) \right) = \int \left(\sum_{\gamma^a \in \Gamma^a} w_\gamma^a h \left(((\Phi^a \omega)^{(1)}, \dots, (\Phi^a \omega)^{(V)}), (\gamma^a(1)\sigma^{(1)}, \dots, \gamma^a(V)\sigma^{(V)}) \right) \right) dP_V(a).$$

3. WHAT IS NEEDED NEXT

- (1) Write down the dual operator on measures in a reasonable notation
- (2) Prove a Ruelle-Perron-Frobenius theorem, (and it may even follow directly from the standard one by defining a transformation on the appropriate space)
- (3) Prove a disintegration result
- (4) Hence get the required measure on a.e. V -variable fractal